Supplementary Material for "Multi-Horizon Failure-Risk Models for Battery Failure Prediction: How Much Cross-Chemistry Transfer Is a State-of-Health Shortcut?"

Hussain Touhid Siddiquee, Syeda Salsabil Islam, Jasimul Islam Chowdhury, and Farzana Hoque Eshica

Department of Electrical and Electronic Engineering, Leading University,
Ragib Nagar, Kamal Bazar, South Surma, Sylhet 3110, Bangladesh

This supplement provides the detailed tables supporting the main paper. All values are computed with the released, version-locked pipeline; pooled AUCs carry cell-level bootstrap 95% confidence intervals except where noted.

S1 Within-Dataset Results on the New Battery Archive Groups

Table S1 reports the horizon-resolved within-dataset AUC for the three tree ensembles on the four Battery Archive groups (Platt-calibrated, cell-disjoint folds). The mostly-censored SNL LFP group (0.031 positive prevalence) is evaluated at $H{=}20$ only and is supporting evidence throughout. Across all groups, 38 of 39 tree–group–horizon configurations reach 0.85, spanning 0.61–1.00.

Table S1: Within-Dataset AUC per Horizon on the Battery Archive Groups (Tree Ensembles, Platt-Calibrated). SNL LFP: $H{=}20$ Only (0.031 Prevalence).

Group	Model	H=10	H=20	H=30	H=50
HNEI NMC	XGBoost	1.000	0.999	0.999	0.999
	LightGBM	0.999	0.999	0.999	0.999
	Random Forest	1.000	1.000	1.000	0.999
SNL NCA	XGBoost	0.994	0.991	0.989	0.980
	LightGBM	0.995	0.992	0.989	0.985
	Random Forest	0.996	0.995	0.993	0.986
SNL NMC	XGBoost	0.872	0.869	0.869	0.878
	LightGBM	0.877	0.887	0.868	0.888
	Random Forest	0.915	0.929	0.888	0.607
SNL LFP	XGBoost	—	0.999	—	—
	LightGBM	—	0.999	—	—
	Random Forest	—	0.999	—	—

On calibration and precision–recall within dataset (NASA, averaged over the three tree ensembles and the four horizons): raw scores reach a mean PR-AUC of 0.774 while isotonic calibration collapses it to 0.639, because the fitted step function creates large tied groups; Platt preserves ranking. Under transfer, isotonic maps fitted on source out-of-fold scores cost the ensembles up to 0.48 pooled AUC on Oxford and 0.31 on Severson even with SOH present.

S2 Monotonicity Violations of Fixed-Horizon Classifiers

Table S2 quantifies how often four independent per-horizon classifiers violate $p_{10} \le p_{20} \le p_{30} \le p_{50}$ on the same scored row, on raw and Platt-calibrated scores. The discrete-time hazard model's survival product is

monotone by construction (rate 0). Raw within-dataset violation rates span 32–45%; transfer inflates them, up to 95.8% for the NASA-trained XGBoost on Severson.

Table S2: Monotonicity Violations: Share of Scored Rows With $p_{10} \leq p_{20} \leq p_{30} \leq p_{50}$ Failing, on Raw (and Platt-Calibrated) Scores. The Survival Product Is Monotone by Construction (Rate 0).

Setting	Target	Model	Raw / Platt
within	calce	XGBoost	0.452 / 0.056
within	nasa	XGBoost	0.323 / 0.260
transfer (calce src)	oxford	LightGBM	0.641 / 0.004
transfer (calce src)	oxford	Random Forest	0.501 / 0.540
transfer (calce src)	oxford	XGBoost	0.565 / 0.003
transfer (nasa+calce src)	oxford	LightGBM	0.210 / 0.648
transfer (nasa+calce src)	oxford	Random Forest	0.335 / 0.305
transfer (nasa+calce src)	oxford	XGBoost	0.609 / 0.341
transfer (nasa src)	oxford	LightGBM	0.521 / 0.468
transfer (nasa src)	oxford	Random Forest	0.559 / 0.376
transfer (nasa src)	oxford	XGBoost	0.248 / 0.124
transfer (calce src)	severson	LightGBM	0.405 / 0.013
transfer (calce src)	severson	XGBoost	0.420 / 0.072
transfer (nasa src)	severson	LightGBM	0.582 / 0.537
transfer (nasa src)	severson	Random Forest	0.778 / 0.557
transfer (nasa src)	severson	XGBoost	0.958 / 0.923
—	—	Survival product	0.000 / 0.000

S3 Discrete-Time Hazard Model: Transfer Results

Table S3 reports transfer at $H=20$ for the survival-product models of Section II-E of the main paper. The formulation inherits the SOH dependence of the independent classifiers: with the ALL-LCO source, hazard XGBoost transfers to Oxford at pooled AUC 0.314, significantly below chance, and the hazard logistic degenerates to exactly 0.500 on three of six pairs.

Table S3: Discrete-Time Hazard Models at $H=20$: Transfer Performance (Pooled AUC With Cell-Level Bootstrap 95% CI).

Training	Model	Oxford	Severson
NASA	Hazard XGBoost	0.971 [0.953,0.991]	0.884 [0.786,0.998]
NASA	Hazard Logistic	0.500 [0.500,0.500]	0.607 [0.402,0.890]
CALCE	Hazard XGBoost	0.466 [0.301,0.578]	0.737 [0.711,0.774]
CALCE	Hazard Logistic	0.500 [0.500,0.500]	0.500 [0.500,0.500]
ALL LCO	Hazard XGBoost	0.314 [0.135,0.457]	0.866 [0.770,0.978]
ALL LCO	Hazard Logistic	0.500 [0.500,0.500]	0.916 [0.873,0.969]

S4 SOH Ablation Evidence (All Horizons and Models)

Table S4 states the central SOH ablation with the cell as the experimental unit, for every ensemble, horizon, and LFP target: pooled AUC with and without SOH, and the paired cell-level bootstrap interval on the difference (primary evidence). On Severson every interval is positive; on Oxford the point estimates agree but paired intervals are undefined with five cells. Cycle-level DeLong p-values underflow throughout and are reported as $p < 10^{-16}$ without interpretation, per the evaluation protocol of the main paper.

Table S4: SOH Ablation (ALL-LCO Source): Pooled AUC With and Without SOH, and Paired Cell-Level Bootstrap 95% CI on the Difference.

Target	**Model**	**H**	**w/ SOH**	**w/o SOH**	**Delta [95% CI]**
Oxford	XGBoost	10	0.917	0.429	(undef.: 5 cells)
	XGBoost	20	0.917	0.429	(undef.: 5 cells)
	XGBoost	30	0.917	0.429	(undef.: 5 cells)
	XGBoost	50	0.917	0.429	(undef.: 5 cells)
	LightGBM	10	0.971	0.332	(undef.: 5 cells)
	LightGBM	20	0.971	0.332	(undef.: 5 cells)
	LightGBM	30	0.971	0.332	(undef.: 5 cells)
	LightGBM	50	0.971	0.332	(undef.: 5 cells)
	Random Forest	10	0.989	0.581	(undef.: 5 cells)
	Random Forest	20	0.989	0.581	(undef.: 5 cells)
	Random Forest	30	0.989	0.581	(undef.: 5 cells)
	Random Forest	50	0.989	0.581	(undef.: 5 cells)
Severson	XGBoost	10	0.896	0.750	+0.137 [0.079,0.249]
	XGBoost	20	0.896	0.750	+0.146 [0.097,0.233]
	XGBoost	30	0.896	0.750	+0.162 [0.115,0.237]
	XGBoost	50	0.896	0.750	+0.180 [0.134,0.250]
	LightGBM	10	0.882	0.718	+0.134 [0.105,0.200]
	LightGBM	20	0.882	0.718	+0.164 [0.127,0.227]
	LightGBM	30	0.882	0.718	+0.102 [0.030,0.230]
	LightGBM	50	0.882	0.718	+0.155 [0.112,0.214]
	Random Forest	10	0.889	0.746	+0.163 [0.133,0.225]
	Random Forest	20	0.889	0.746	+0.143 [0.106,0.226]
	Random Forest	30	0.889	0.746	+0.171 [0.138,0.232]
	Random Forest	50	0.889	0.746	+0.175 [0.114,0.264]

S5 GRU Transfer Across Seeds

Table S5 reports the three-seed GRU transfer to Severson at $H=20$ on the common four-feature subset (per-cell mean AUC $\pm$ across-seed SD). Without SOH the CALCE-trained GRU is seed-unstable — individual seeds span 0.21–1.00 — so sequence representations do not reproducibly rescue transfer.

Table S5: GRU Transfer to Severson at $H=20$ (Common Feature Subset, Per-Cell Mean AUC $\pm$ Across-Seed SD, Three Seeds).

Training	With SOH	Without SOH
NASA	0.998±0.003	0.987±0.015
CALCE	0.964±0.033	0.479±0.451
ALL LCO	0.991±0.015	0.893±0.131

S6 Small-Sample Recalibration (Arms A and B)

Tables S6 and S7 detail the two recalibration arms summarized in Section III-H of the main paper, on Severson without SOH at $H=20$. Arm A refits only the calibrator on k labeled target cells; Arm B continues training the classifier. Arm A repairs calibration error but not discrimination; Arm B's recovery is source-dependent, and its ratio above 1.0 for the NASA-trained XGBoost is a ceiling-mismatch artifact, not super-ceiling recovery.

Table S6: Arm A Recalibration on Severson Without SOH ($H=20$): Values Average the Three LCO Sources and Available Tree Models; ECE Uses 10 Equal-Width Probability Bins.

k	Zero ECE	Iso. ECE	Platt ECE	Temp. ECE	Iso. AUC	Platt AUC	Temp. AUC
5	0.817	0.044	0.045	0.563	0.616	0.599	0.669
10	0.817	0.010	0.010	0.561	0.634	0.646	0.651
20	0.817	0.013	0.013	0.560	0.634	0.655	0.648
40	0.817	0.015	0.015	0.561	0.637	0.657	0.653

Table S7: Arm B Recovery at $k=5$ on Severson Without SOH, $H=20$. Recovery Ratio Uses the Within-LCO Ceiling; Retention Is the LCO-Holdout AUC Before and After the Update.

Training	Model	Zero AUC	Arm B AUC	Recovery	DeLong p	LCO retention
NASA	XGBoost	0.620	0.854	$+1.04^{*}$	$< 10^{-3}$	$1.000 \rightarrow 1.000$
NASA	LightGBM	0.493	0.838	$+0.88$	$< 10^{-3}$	$1.000 \rightarrow 0.997$
NASA	Random Forest	0.724	0.751	$+0.19$	$< 10^{-3}$	$0.959 \rightarrow 0.314$
CALCE	XGBoost	0.686	0.796	$+0.53$	$< 10^{-3}$	$0.997 \rightarrow 0.996$
ALL LCO	XGBoost	0.760	0.828	$+0.47$	$< 10^{-3}$	$0.690 \rightarrow 0.567$

*Ratio above 1.0 is a ceiling-mismatch artifact, not super-ceiling recovery.

S7 Factorial Feature Ablation Grid

Table S8 reports the full $2^3 - 1$ factorial grid over feature groups {SOH, cycle index, sensors} at $H=20$ (XGBoost, pooled AUC with cell-level bootstrap 95% CI; sensors = voltages, current, temperature, duration). No combination without SOH reaches 0.76 on either target (best 0.75); removing both SOH and the cycle index leaves essentially nothing.

Table S8: Factorial Feature Ablation at $H=20$ (XGBoost, Pooled AUC With Cell-Level Bootstrap 95% CI).

	Oxford			Severson		
Feature set	**NASA src**	**CALCE src**	**ALL src**	**NASA src**	**CALCE src**	**ALL src**
full	0.957	0.850	0.917	0.871	0.850	0.896
no SOH	0.518	0.416	0.429	0.595	0.682	0.750
−cycle	0.965	0.967	0.993	0.854	0.766	0.774
−SOH−cycle	0.658	0.433	0.388	0.531	0.500	0.535
SOH only	0.965	1.000	1.000	0.834	0.798	0.833
cycle only	0.541	0.541	0.541	0.709	0.694	0.719
sensors	0.658	0.433	0.388	0.531	0.500	0.535

S8 Operating-Condition Shift at Fixed Chemistry and Laboratory

Table S9 details the leave-condition-out protocol summarized in Section III-M of the main paper: the test cell is drawn from an operating condition (temperature band, C-rate, or both) that contributes no training cell in that fold, and condition metadata never enters the features. The unfitted SOH distance rule is essentially undisturbed (0.886–0.999). The mixed-fold reference on the 6-cell SNL NMC group is unstable (raw AUC 0.65 with SOH, against the 0.87 reached under the main paper's cell-disjoint outer folds), so its reference columns read "—" and no drop is claimed. HNEI is omitted (single condition); SNL LFP was run with the with-SOH and common no-SOH feature sets only.

Table S9: Leave-Condition-Out Versus Mixed-Cell Folds at $H=20$ (XGBoost, Pooled AUC). Ref. Is the Ordinary Mixed-Cell AUC for the Same Feature Set (— Where Unstable or Not Run); the Distance Rule Is the Unfitted $0.80-$SOH Score.

		With SOH			No SOH		
Group	**Held-out axis**	**LCO**	**Dist. rule**	**Ref.**	**LCO**	**Dist. rule**	**Ref.**
SNL NCA	temp $imes$ rate	0.989	0.987	0.985	0.894	0.987	0.928
SNL NCA	temperature	0.989	0.987	0.985	0.888	0.987	0.928
SNL NCA	C-rate	0.961	0.987	0.985	0.820	0.987	0.928
SNL NMC	temp $imes$ rate	0.847	0.886	—	0.666	0.886	—
SNL NMC	temperature	0.787	0.886	—	0.569	0.886	—
SNL NMC	C-rate	0.847	0.886	—	0.666	0.886	—
SNL LFP	temp $imes$ rate	0.991	0.996	0.994	—	—	—
SNL LFP	temperature	0.995	0.999	0.994	—	—	—
SNL LFP	C-rate	0.994	0.998	0.994	—	—	—

S9 Same-Chemistry Controls With the New Groups as Sources

Table S10 extends the same-chemistry cross-laboratory controls of Section III-F of the main paper to the new groups as *sources*: NMC→NMC and NCA/NMC cross-laboratory pairs within the SNL matrix, and LFP→LFP transfer from SNL LFP to Oxford and Severson. The pattern repeats: high transfer with SOH, near-chance sensor-only transfer on the legacy LFP targets.

Table S10: Same-Chemistry Controls With New Groups as Sources at $H=20$ (XGBoost, Pooled AUC). Common = Shared Four-Feature Subset.

Pair	**with SOH**	**no SOH**	**no SOH (common)**
HNEI NMC → SNL NMC (NMC)	0.868	0.753	0.756
SNL NCA → SNL NMC (NCA/NMC)	0.905	0.801	0.752
SNL LFP → Oxford (LFP)	1.000	0.857	0.857
SNL LFP → Severson (LFP)	0.820	0.356	0.394